

Lecture 20

Game Theory: Game-Theoretic Foundations for Multi-Agent Learning

COMP532 – Machine Learning and Bioinspired Optimisation

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Today's Questions

At the end of this lecture, you should be able to answer:

1. Why is game theory a natural language for multi-agent learning?
2. What is a normal-form game, and how does it model strategic interaction?
3. What are best responses, dominant strategies, and Nash equilibria?
4. Why do we need stochastic / Markov games beyond matrix games?

Recap: Why Standard RL is Not Enough

- ▶ In single-agent RL, the environment is typically treated as stationary.
- ▶ In multi-agent learning, other agents are part of the environment.
- ▶ If other agents adapt, then the effective environment changes over time.

Key issue

An agent must reason not only about **states** and **rewards**, but also about **the actions of other decision-makers**.

This is precisely the setting studied by **game theory**.

Why Game Theory?

Reinforcement Learning gives:

- ▶ learning from interaction
- ▶ value functions
- ▶ policies and updates
- ▶ sequential optimisation

Game Theory gives:

- ▶ formal models of strategic interaction
- ▶ best response reasoning
- ▶ equilibrium concepts
- ▶ tools for cooperation and competition

Main message

To study multi-agent learning, we need both:

learning + strategic reasoning.

A First Strategic Example

		Agent 2	
		A	B
Agent 1	A	(3,3)	(0,1)
	B	(1,0)	(2,2)

- ▶ Agent 1 chooses a row; Agent 2 chooses a column.
- ▶ Each cell contains a payoff pair:

(payoff to Agent 1, payoff to Agent 2)

- ▶ Example: if Agent 1 plays **A** and Agent 2 plays **B**, the result is (0, 1).

Strategic dependence

There is no single best action independently of the other agent's choice.

Normal-Form Games

A **normal-form game** (or matrix game) consists of:

$$\mathcal{G} = \langle N, (\mathcal{A}_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

where:

- ▶ $N = \{1, \dots, n\}$ is the set of players
- ▶ $\mathcal{A}_i$ is the action set of player i
- ▶ $u_i(a_1, \dots, a_n)$ is the payoff of player i

Interpretation

Each player's payoff depends on the **joint action**, not just its own action.

Why Normal-Form Games Matter for MARL

- ▶ They capture the simplest form of strategic interaction.
- ▶ They isolate the effect of **mutual dependence** in decision-making.
- ▶ They are the building block for more complex sequential settings.

Teaching intuition

Before studying multi-agent learning in many states, it helps to understand the single-state case extremely well.

Matrix games $\longrightarrow$ Stochastic / Markov games

Best Response

Suppose player i believes that the other players will choose actions a_{-i} .

A **best response** is an action a_i^* such that

$$a_i^* \in \arg \max_{a_i \in \mathcal{A}_i} u_i(a_i, a_{-i}).$$

- ▶ A best response depends on what others are expected to do.
- ▶ Best response is therefore a **conditional** notion.

Key intuition

Strategic reasoning means choosing the best action *relative to others' behaviour*.

Best Response in the Example

	A	B
A	(3,3)	(0,1)
B	(1,0)	(2,2)

For player 1:

- ▶ If player 2 plays **A**, then player 1 prefers **A** since $3 > 1$.
- ▶ If player 2 plays **B**, then player 1 prefers **B** since $2 > 0$.

Similarly for player 2:

- ▶ Best response to **A** is **A**.
- ▶ Best response to **B** is **B**.

Both players want to match the other player's action.

Extending to 3 agents

Suppose agents 1, 2, and 3 each choose **A** or **B**.

$$u_i(a_1, a_2, a_3) = \begin{cases} 3 & \text{if all three actions match} \\ 0 & \text{otherwise} \end{cases}$$

Examples:

- ▶ $(A, A, A) \rightarrow (3, 3, 3)$
- ▶ $(B, B, B) \rightarrow (3, 3, 3)$
- ▶ $(A, A, B) \rightarrow (0, 0, 0)$

Now each agent's best response depends on the joint actions of the other two agents.

Dominant Strategy

An action a_i^* is a **dominant strategy** for player i if

$$u_i(a_i^*, a_{-i}) \geq u_i(a_i, a_{-i}) \quad \text{for all } a_i \in \mathcal{A}_i \text{ and all } a_{-i}.$$

- ▶ A dominant strategy is best **regardless** of what others do.
- ▶ This is stronger than best response.

Important distinction

Best response: best given others' actions.

Dominant strategy: best no matter what others do.

Example: Prisoner's Dilemma

	Cooperate	Defect
Cooperate	$(-1, -1)$	$(-3, 0)$
Defect	$(0, -3)$	$(-2, -2)$

For each player:

- ▶ If the other cooperates, defect gives a better payoff
- ▶ If the other defects, defect still gives a better payoff

Conclusion

Defect is a dominant strategy for both players.

Why Prisoner's Dilemma Matters

- ▶ Individual rationality leads both players to defect.
- ▶ Yet mutual cooperation would produce a better joint outcome.

Core tension

What is good for an individual can be bad for the group.

This example introduces a recurring theme in multi-agent learning:

- ▶ competition vs cooperation
- ▶ self-interest vs social welfare
- ▶ local incentives vs global outcomes

Nash Equilibrium

A joint action $a^* = (a_1^*, \dots, a_n^*)$ is a **Nash equilibrium** if, for every player i ,

$$u_i(a_i^*, a_{-i}^*) \geq u_i(a_i, a_{-i}^*) \quad \text{for all } a_i \in \mathcal{A}_i.$$

Interpretation

A Nash equilibrium is a situation in which each player's action is a best response to the actions of the others.

At a Nash equilibrium, no player can improve its payoff by changing its action *unilaterally*.

Equivalent intuition

If all other players keep their actions fixed, no single player wants to deviate on their own.

Nash equilibrium is a notion of **strategic stability**.

Nash Equilibrium in the Coordination Example

	A	B
A	(3,3)	(0,1)
B	(1,0)	(2,2)

The action profiles:

- ▶ (A, A) is a Nash equilibrium
- ▶ (B, B) is also a Nash equilibrium

Why?

- ▶ At (A, A) , neither player wants to deviate alone
- ▶ At (B, B) , neither player wants to deviate alone

Key insight

Games may have **multiple equilibria**, which creates an equilibrium selection problem.

Static Equilibrium vs Learning

Nash equilibrium tells us what a stable joint action profile looks like.

But it does **not** tell us:

- ▶ how agents discover that equilibrium
- ▶ whether learning converges to it
- ▶ which equilibrium is selected when several exist
- ▶ what happens when agents keep adapting online

This is the bridge to learning

Game theory gives a **solution concept**; multi-agent learning asks how agents might actually **reach** or **fail to reach** such solutions.

Mixed Strategies

In some games, deterministic actions may be too predictable.

A **mixed strategy** is a probability distribution over actions.

If player i chooses action a_i with probability $\pi_i(a_i)$, then the payoff becomes an expected payoff under the induced distribution over joint actions.

Why mixed strategies matter

They allow equilibrium reasoning in settings where pure strategies are insufficient.

Example: Matching Pennies

	Heads	Tails
Heads	(1,-1)	(-1,1)
Tails	(-1,1)	(1,-1)

- ▶ Player 1 wants to **match** the actions.
- ▶ Player 2 wants to **mismatch** the actions.
- ▶ No pure action is stable:
 - ▶ if player 1 chooses **Heads**, player 2 prefers **Tails**;
 - ▶ if player 1 chooses **Tails**, player 2 prefers **Heads**.

Why mixed strategies?

To avoid being predictable, each player can randomise:

$$\pi(\text{Heads}) = 0.5, \quad \pi(\text{Tails}) = 0.5$$

This means choosing **Heads** with probability 0.5 and **Tails** with probability 0.5.

Zero-Sum vs General-Sum Games

Zero-sum games

- ▶ One player's gain is the other's loss
- ▶ $u_1 + u_2 = 0$
- ▶ Pure competition
- ▶ Example: **Chess**, Poker, Matching Pennies

Typical applications:

- ▶ Adversarial planning
- ▶ Security games
- ▶ Competitive board games

Practical importance

Many multi-agent learning algorithms behave very differently in adversarial settings than in mixed cooperative–competitive environments.

General-sum games

- ▶ Payoffs need not cancel out
- ▶ Cooperation and conflict may coexist
- ▶ More realistic for many applications
- ▶ Example: **Driving**, congestion control, trading

Typical applications:

- ▶ Multi-robot coordination
- ▶ Autonomous driving
- ▶ Resource allocation

Limits of Matrix Games

Normal-form games are very useful, but they are limited:

- ▶ They describe **single-shot** or single-state interaction
- ▶ They do not explicitly model state transitions
- ▶ They do not capture long-term consequences over trajectories

What is missing?

In many real systems, agents interact **repeatedly** in an evolving environment.

We therefore need a sequential generalisation.

From Matrix Games to Stochastic Games

A **stochastic game** (or Markov game) extends an MDP to multiple agents.

It includes:

- ▶ a state space $\mathcal{S}$
- ▶ action spaces $\mathcal{A}_1, \dots, \mathcal{A}_n$
- ▶ a transition function

$$P(s' \mid s, a_1, \dots, a_n)$$

- ▶ reward functions

$$r_i(s, a_1, \dots, a_n)$$

Core difference from MDPs

Both rewards and transitions can depend on the **joint action**.

Why Markov Games Matter

- ▶ They combine **sequential decision-making** with **strategic interaction**.
- ▶ They are the natural model for multi-agent RL in many-state environments.
- ▶ They generalise both:
 - ▶ MDPs (single-agent case)
 - ▶ matrix games (single-state case)

MDP $\subset$ Markov Game

Matrix Game $\subset$ Markov Game

Bridge forward

Once we formalise the environment as a Markov game, we can ask: How should agents learn in it?

A Conceptual Comparison

Setting	Agents	States	Strategic interaction
Bandit / single-state RL	1	1	No
MDP	1	Many	No
Matrix game	Many	1	Yes
Markov game	Many	Many	Yes

Mental map

Matrix games help us understand strategic dependence.

Markov games let us study strategic dependence over time.

Summary

1. Game theory formalises **strategic interaction**.
2. In normal-form games, payoffs depend on the **joint action**.
3. Best response, dominant strategy, and Nash equilibrium are central solution concepts.
4. Matrix games are useful but limited: they do not model sequential interaction.
5. Markov games generalise MDPs to multi-agent settings.

Looking Ahead

Next lecture:

- ▶ How do agents adapt over repeated interaction?
- ▶ Does equilibrium analysis explain learning dynamics?
- ▶ What happens when players update policies over time?

Bridge to Lecture 21

Equilibrium is a static notion of stability.

The next question is dynamic: **how do agents learn?**